

Array

Syntax

`dataType[] arrayName = new dataType[size];`

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[Easy to manage, loop, & process]

↳ An array is a collection of elements of the same data type

↳ it allow you to store multiple values of the same type in a single variable.

(with array)

(without array)

Ex `int[] marks = {85, 90, 75, 95, 88};`

`int mark1 = 85;`

(array size is fixed) ↳ elements must be of same type in array

[It also called as linear form]

Types

↳ Easy access

↳ iteration

↳ Memory efficiency

↳ Code Simplicity

↳ searching / sorting.

* One-Dimensional array

→ [A single row of elements]

* Two-Dimensional array

→ [(Matrix) An arrays of arrays - used to represent tables]

```
int [][] matrix = {  
    {1, 2, 3},  
    {4, 5, 6},  
    {7, 8, 9}.  
};
```

```
int [] arr = {1, 2, 3, 4, 5};
```

One-dimensional array.

Tagged array

Syntax

`dataType[][] arrayName = new dataType[numberOfRows][];`

`arrayName[0] = new dataType[size];`

`arrayName[1] = new dataType[size];`

`arrayName[2] = new dataType[size];`

→ {each row with different column sizes}

(array of arrays)

* Tagged array is a two dimensional array where each row can have a different number of columns.

* Normal 2D arrays $\rightarrow$ all rows have same length

* Jagged Array $\rightarrow$ each row can have a different length.

ex: Row 0 $\rightarrow$ 10, 20
Row 1 $\rightarrow$ 30, 40, 50
Row 2 $\rightarrow$ 60.

```
[  
    [10, 20],  
    [30, 40, 50],  
    [60]  
]
```